Automated Mineralogy using SEM-EDS Images Reveals Diverse Equilibrated Inclusions

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Motivation

- Modern instrumentation allows the collection of very large data volumes of chemical microanalytical data in the form of X-ray chemical maps using scanning electron microscopes equipped for **energy dispersive X-ray spectroscopy** (EDS). A typical EDS data set for a petrographic thin section might produce ten or more images, imaging an area of several cm² at a resolution of a few µm/pixel. It is rapidly becoming impractical for a human investigator to carefully examine the full dataset. Reliable automated classification can form the basis of further qualitative and quantitative analyses, particularly when the classes can be assigned physical meaning in terms of mineral phases.

Methods

- Mineral compositional maps (22 EDS channels + BSE) were acquired from a carbon-coated thin section of NWA 4587 using a MIRA3 SEM at 25 kV and 8.24 nA with 3 µm pixel spacing. The raw color mapped EDS map PNGs were converted to scalar [0,1] intensities via nearest-neighbor lookup table inversion. A valid-region polygon mask and element-zero background masking isolated ~10.7M mineral pixels. Bilateral denoising preserved grain boundaries, followed by per-channel z-score normalization and PCA dimensionality reduction (auto-selected at 95% cumulative variance).
- Clustering was performed using a tiled progressive HDBSCAN strategy: the image was divided into 1024×1024 px tiles, each independently clustered, with phases merged across tiles via a cosine-similarity registry (threshold 0.88). A two-pass workflow first identified major phases, then reclustered unassigned pixels with relaxed parameters for rare phase discovery. Final kNN noise reassignment (k=5, distance-weighted voting) resolved boundary pixels. The entire workflow is automated in Python using SciPy, scikit-learn, and hdbscan, requiring no proprietary software.

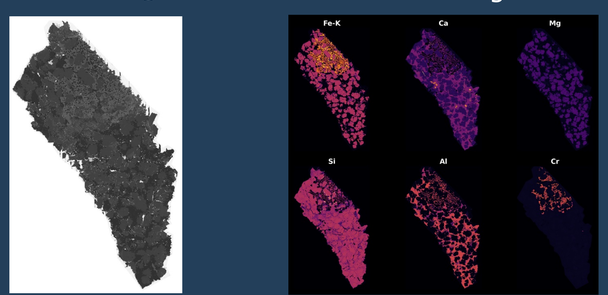
Results

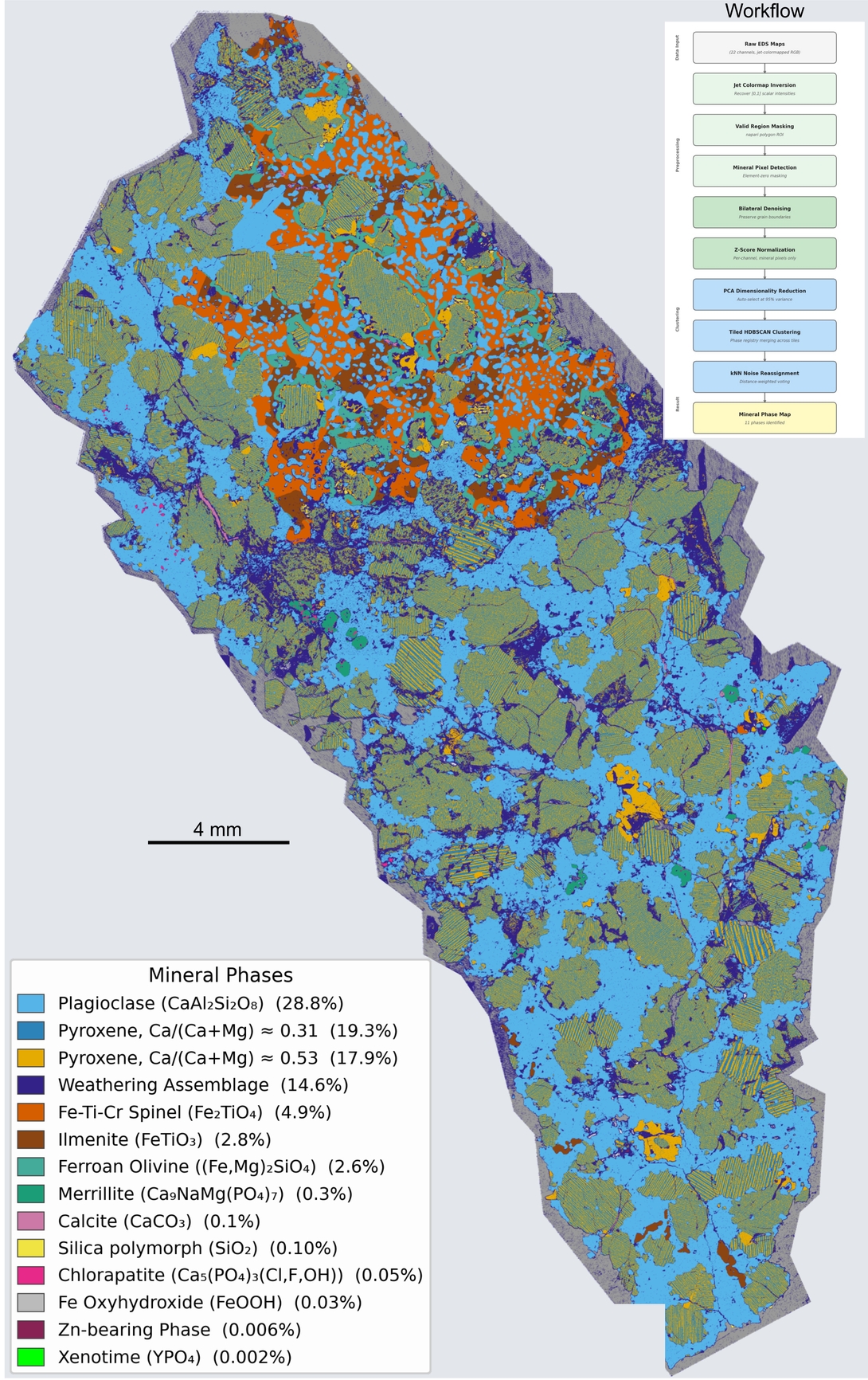
- The workflow identified 11 distinct mineral phases from 12.5M mineral pixels. The dominant phases — low-Ca pyroxene (54.4%) and calcic plagioclase (28.8%) — are consistent with the known petrology of NWA 4587. Accessory phases including ilmenite (2.8%), Fe-Ti-Cr spinel (4.9%), merrillite (0.4%), apatite (0.1%), and silica (0.1%) were correctly separated despite comprising small modal fractions. Notably, a 213-pixel Y-phosphate grain (xenotime) was recovered in the rare-phase pass, demonstrating the method's sensitivity to volumetrically minor phases.
- Known EDS spectral overlaps (Ba Lα/Ti Kα, Sb Lα/Ca Kα) did not impede phase discrimination — HDBSCAN implicitly handles correlated channels by clustering in the full feature space. One phase (8.4%) remains unidentified, with uniformly low EDS signal but high BSE intensity, possibly representing an Fe-oxyhydroxide-rich terrestrial weathering product.

NWA 4587 Image Data

- Northwest Africa 4587 is an ungrouped achondrite paired with NWA 011. Isotopic systematics of Cr, Ti, and Fe have shown that NWA 4587 is associated with the "carbonaceous" or outer solar system reservoir of known planetary materials, a property thus far unique among known basaltic achondrites[1].

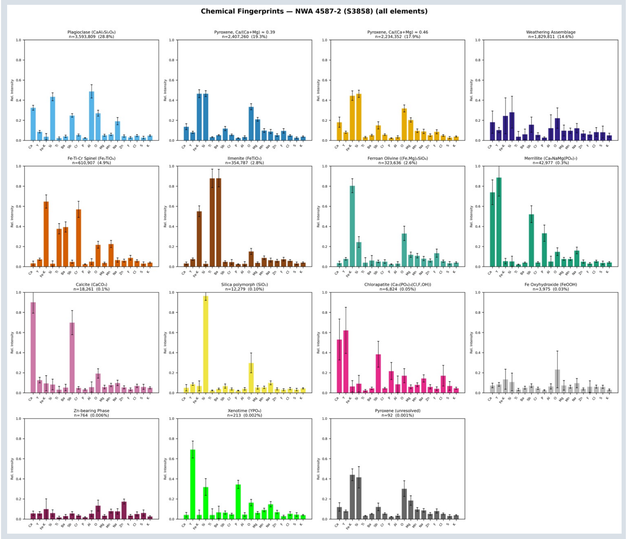
BSE Image + EDS Images





Workflow

Phase Fingerprints



Xenotime is a rare yttrium phosphate comprising just 0.002% of mineral pixels in this section. Its identification by EDS is complicated by the severe spectral overlap between Y-L and P-K lines at ~2.0 keV, which can obscure both elements. Despite these challenges, unsupervised clustering isolated a coherent grain with strongly correlated Y and P signals and as BSE brightness consistent with a high mean atomic number. This demonstrates the pipeline's sensitivity to trace phases without prior mineralogical assumptions.

Conclusions

- Automated HDBSCAN clustering of SEM-EDS image stacks successfully recovers both major rock-forming minerals and rare accessory phases and inclusions in NWA 4587 without manual training data or proprietary software.
- The tiled progressive strategy with phase registry merging handles large images (>10M pixels) within practical memory constraints while maintaining phase continuity across tile boundaries.
- Known EDS spectral overlaps do not prevent correct phase discrimination when clustering in the full multi-element feature space.
- Unusual and unexpected phases (e.g., xenotime) were identified. Even if the xenotime proves to be terrestrial contamination, the fact that our approach identified it demonstrates the power of the method.
- These methods are applicable to any existing chemical imaging dataset and provide a basis for automated granulometry and modal analysis. In the future, data types can be incorporated as well.

References

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